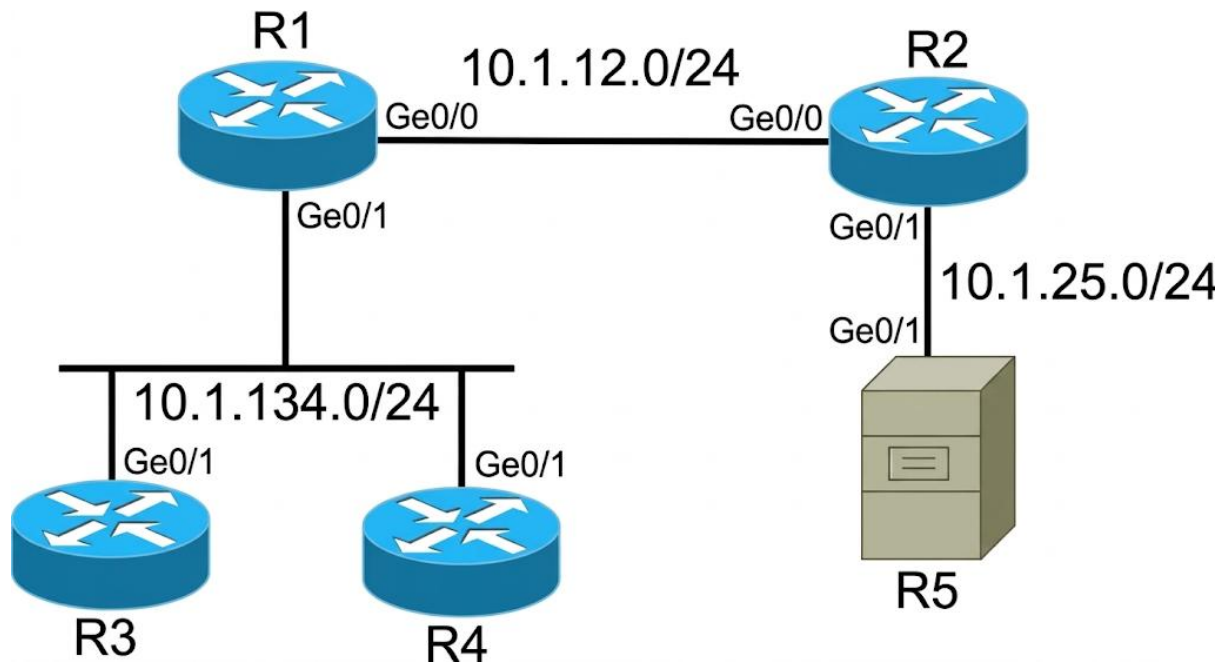

Dynamic NAT



Objective

Configure Dynamic NAT to pool global IP addresses and assign them dynamically to internal hosts requesting external access.

Tasks

- Configure hostnames and IP addresses across all devices.
- Disable routing on **R3** and **R4**; set default gateway to **10.1.134.1**.
- Disable routing on **R5**; set default gateway to **10.1.25.2**.
- Configure default routing on **R1** pointing to **10.1.12.2**.
- Enable HTTP service on **R5**.
- Configure Dynamic NAT on **R1** for the internal subnet **10.1.134.0/24**.
- Verify connectivity from **R3** and **R4** to **R5**.

Concept & Overview

Dynamic NAT maps a group of private real addresses to a defined pool of public addresses on a first-come, first-served basis. Implementation requires:

1. An **Access Control List (ACL)** defining internal traffic permitted for translation.
2. A **NAT Pool** specifying available public IPv4 addresses.
3. A **Binding Statement** coupling the ACL to the pool.

Unlike Static NAT, translation entries do not exist in advance; they are dynamically created when outbound traffic hits the NAT inside interface and expire after idle timeouts.

Device Configurations

Router R1

```
enable
```

```
configure terminal
```

```
hostname R1
```

```
!
```

```
interface GigabitEthernet0/1
```

```
ip address 10.1.134.1 255.255.255.0
```

```
ip nat inside
```

```
!
```

```
interface GigabitEthernet0/0
```

```
ip address 10.1.12.1 255.255.255.0
```

```
ip nat outside
```

```
!
```

```
ip route 0.0.0.0 0.0.0.0 10.1.12.2
```

```
!
```

```
access-list 1 permit 10.1.134.0 0.0.0.255
```

```
!
```

```
ip nat pool TRANSLATE 10.1.12.3 10.1.12.254 prefix-length 24
```

```
ip nat inside source list 1 pool TRANSLATE
```

```
!
```

```
end
```

(Configurations for R2, R3, R4, and R5 remain identical to Lab 11).

Verification

Initiate traffic from **R3** (ICMP) and **R4** (Telnet) towards **R5**:
R3#ping 10.1.25.5

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.1.25.5, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 108/120/132 ms

R4#telnet 10.1.25.5

Trying 10.1.25.5 ... Open

User Access Verification

Password:

R5>

Verify dynamic allocation in **R1**'s translation table:R1#show ip nat translations

Pro	Inside global	Inside local	Outside local	Outside global
icmp	10.1.12.3:16	10.1.134.3:16	10.1.25.5:16	10.1.25.5:16
---	10.1.12.3	10.1.134.3	---	---
tcp	10.1.12.4:29511	10.1.134.4:29511	10.1.25.5:23	10.1.25.5:23
---	10.1.12.4	10.1.134.4	---	---

Because **R3** initiated traffic first, it received the first dynamic pool address (**10.1.12.3**). **R4** requested access second and was assigned **10.1.12.4**.